

# Tracking Consciousness with Median Nerve Stimulation-based BCI during General Anesthesia

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Through EEG and BCI



COPENHAGEN

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# Accidental Awareness during General Anesthesia (AAGA)



Yuko Shimizu

# Monitors to avoid Accidental Awareness during General Anesthesia

**PROBLEM:** **Awareness still occurs** with current monitors [1, 2, 3, 4, 5] : indices were developed from adult cohorts [5], might be falsely increased with EMG activity [3, 4] and are less accurate with age [5]

**OUR SOLUTION:** Actively analyze EEG responses to **Median Nerve Stimulation** (MNS) [6]



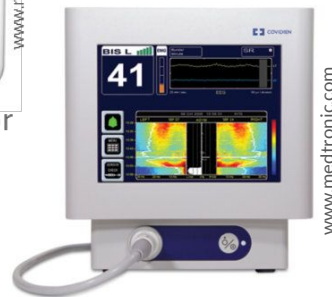
NeuroSENSE monitor



SedLine monitor



Narcotrend monitor



Bispectral Index monitor



Thomas Hendry, Neuromod, 2024

[1] Bruhn et al. Anesth, 2000.

[2] Myles et al. Lancet, 2004.

[3] Sebel et al. Anesth. Analg. 2004.

[4] Aho et al. Acta Anaesthesiol Scand, 2012.

[5] Laferrière-Langlois et al. Anesth. Analg., 2024.

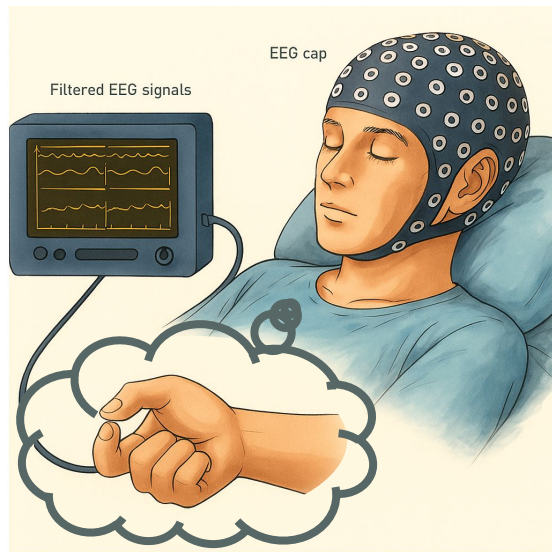
[6] Rimbart et al, Front. Neurosci, 2019.

# I. Median Nerve Stimulation-based BCI to detect motor intentions

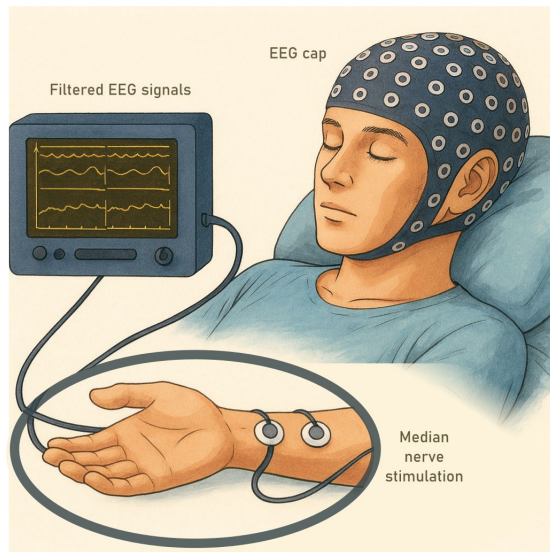
# Data collection: CHRU de Nancy, France



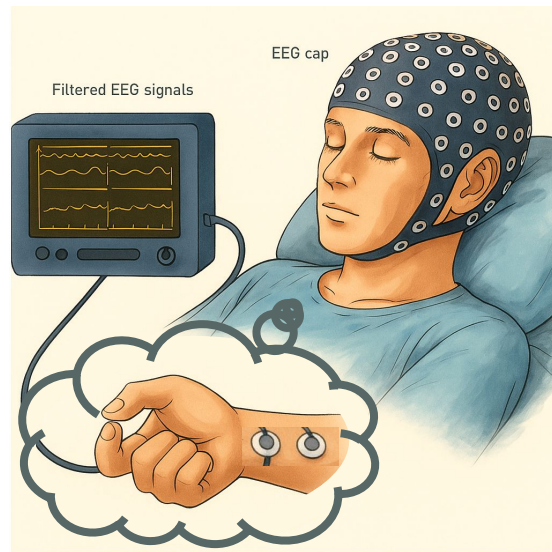
31 subjects, 51 trials of MI, MNS or MNS+MI at 3 concentrations of propofol sedation [14].



MOTOR IMAGERY (MI)



MEDIAN NERVE STIMULATION (MNS)



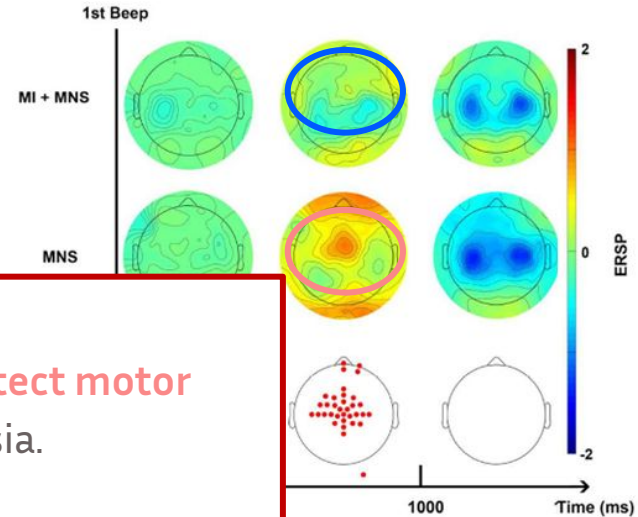
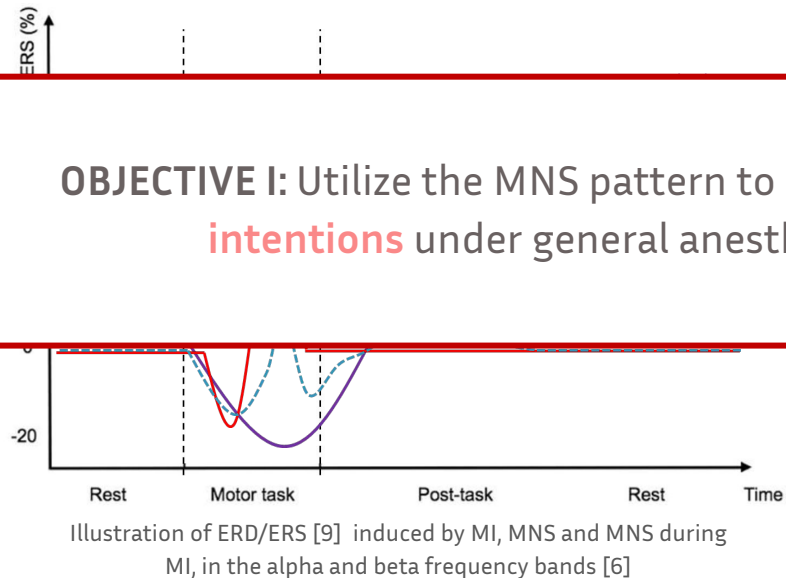
MNS + MI

# I. Median Nerve Stimulation-based BCI to detect motor intentions

**MNS-based BCI** : MNS sensorimotor modulations change with concomitant MI [7, 8], leading to better MI detection accuracy (+12%) [6].

$$ERD/ERS\% = \frac{\overline{x^2} - \overline{BL^2}}{\overline{BL^2}} \times 100$$

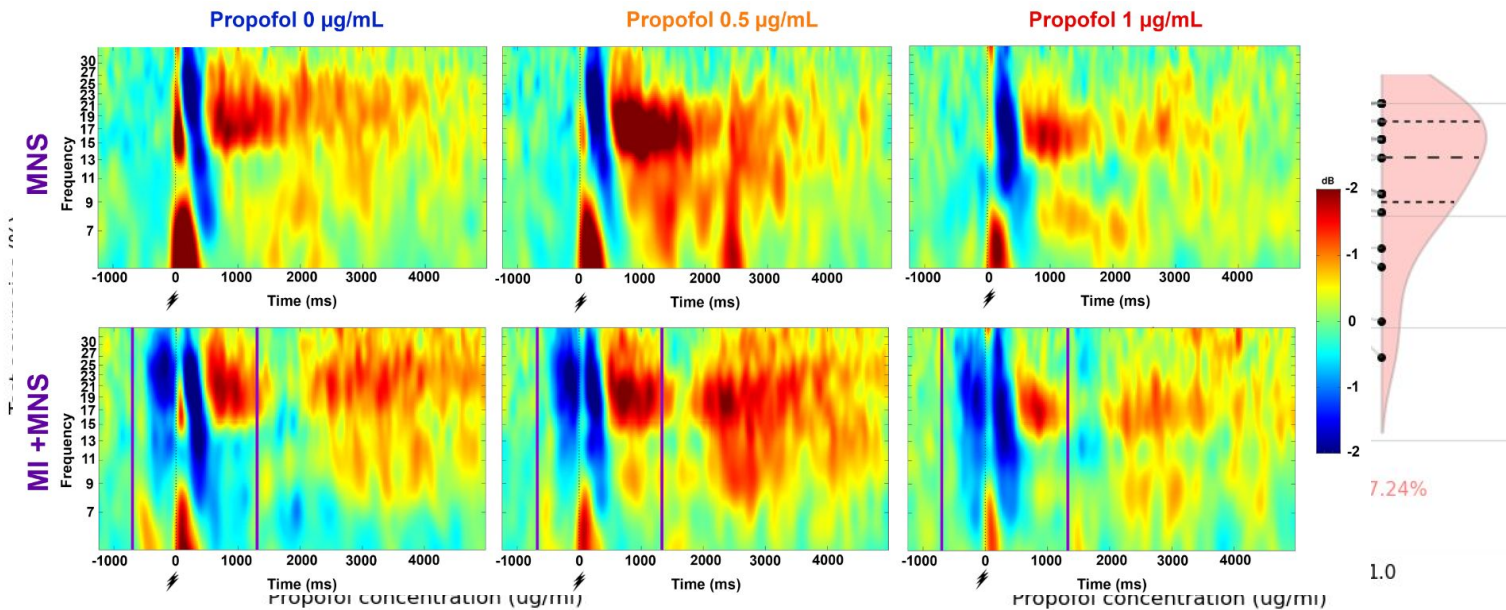
**OBJECTIVE I:** Utilize the MNS pattern to **detect motor intentions** under general anesthesia.



Topographic map of ERD/ERS% (grand average, n = 16) in 8-30 Hz [6]

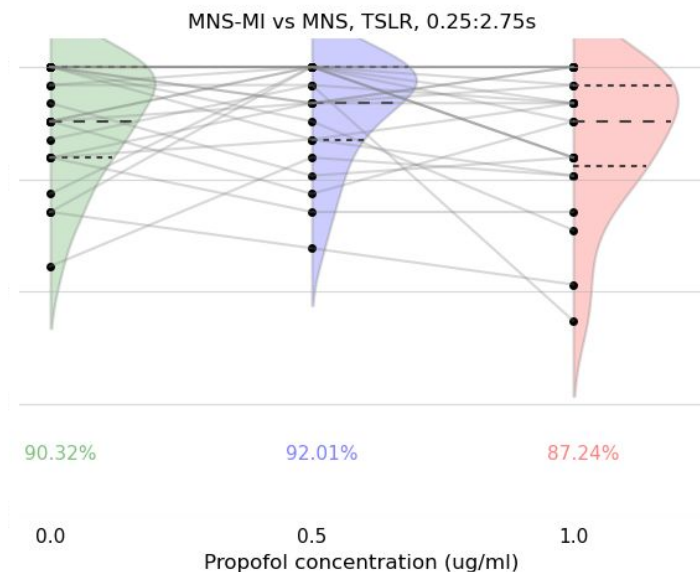
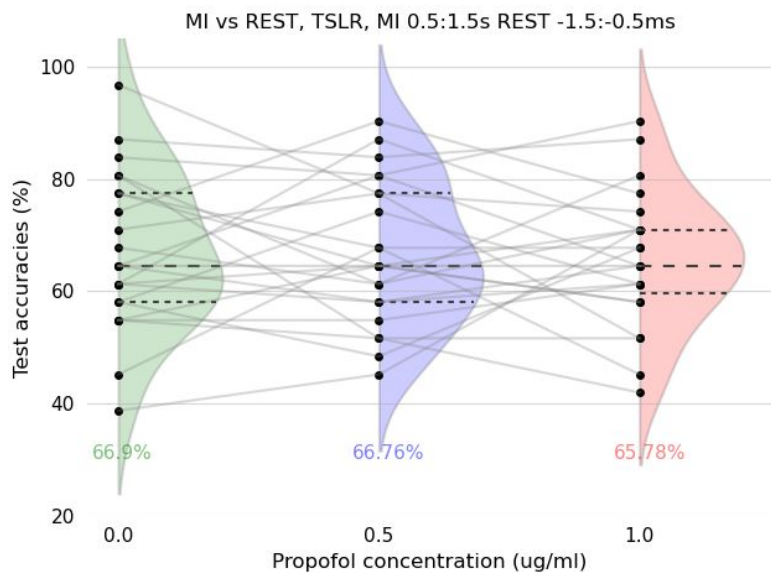
- [6] Rimbert et al, Front. Neurosci, 2019.
- [7] Cheron et al, Electroenceph. Clin. Neuroph., 1987.
- [8] Salenius et al, NeuroImage, 1996.
- [9] Pfurtscheller and Lopes da Silva, Clin Neuroph, 1999.

# Detection of motor imagery under sedation



Time frequency maps (grand average,  $n = 16$ ) in 8-30 Hz, for electrode C3 at different propofol concentrations [6]

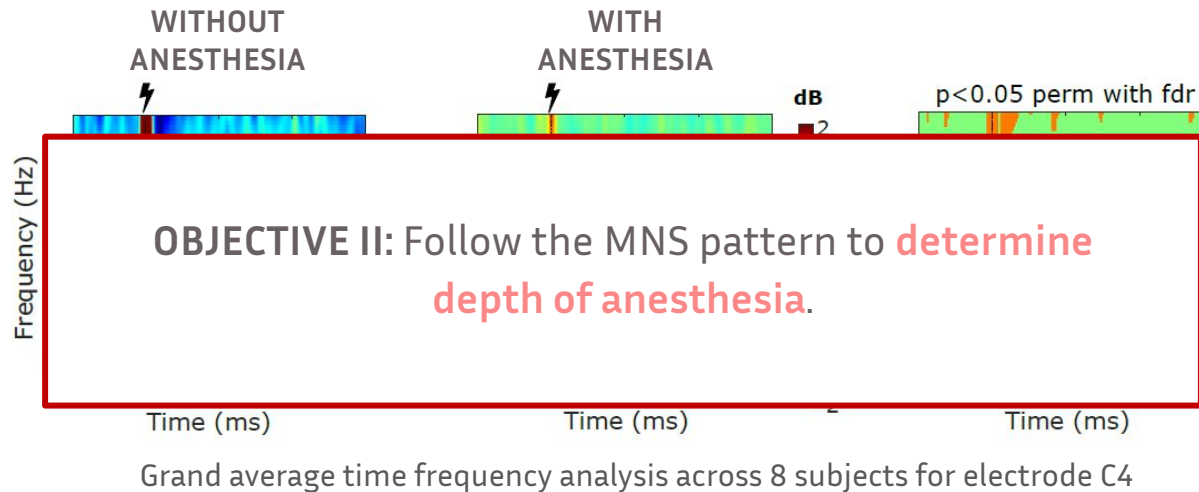
# Detection of motor imagery under sedation



## II. Median Nerve Stimulation-based BCI to monitor anesthesia

## II. Median Nerve Stimulation-based BCI to monitor anesthesia

MNS induces sensorimotor modulations visible via EEG and variable with the depth of anesthesia [7].



PSR: Post-Stimulation Rebound  
ERD: Event Related Desynchronization  
PMBR: Post-Movement Beta Rebound

[7] Marissens Cueva et al. 9th Int.. Graz BCI Conf. 2024.

# Data collection: clinical protocol at CHU Brugmann

14 patients: 8 females;  $50 \pm 14.3$  years old [8]

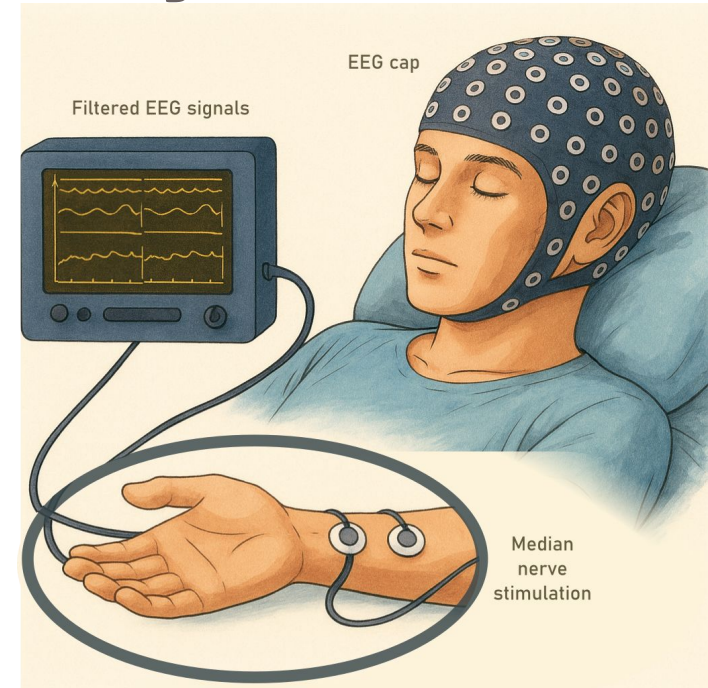


Figure inspired by Sigl and Chamoun, J Clin Monit, 1994

# EEG recording difficulties in a clinical environment

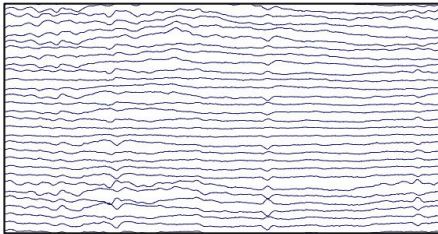
Inclusion of subjects:

- consentment
- position during surgery
- unmaintained surgery
- dysfunctional equipment

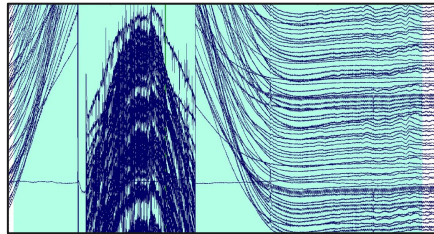


Interferences:

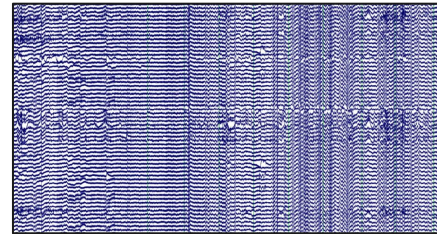
- hot air blankets [9]
- electrocautery [10]
- individual socket
- line noise (50 Hz or 60 Hz)
- movement



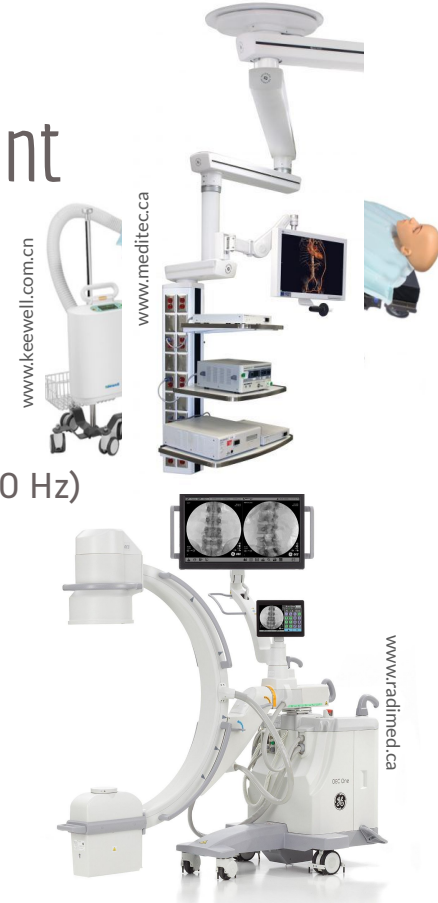
CLEAN



ELECTROCAUTERY



LINE



[9] Hemmerling et al. Anesth. Analg. 2002.

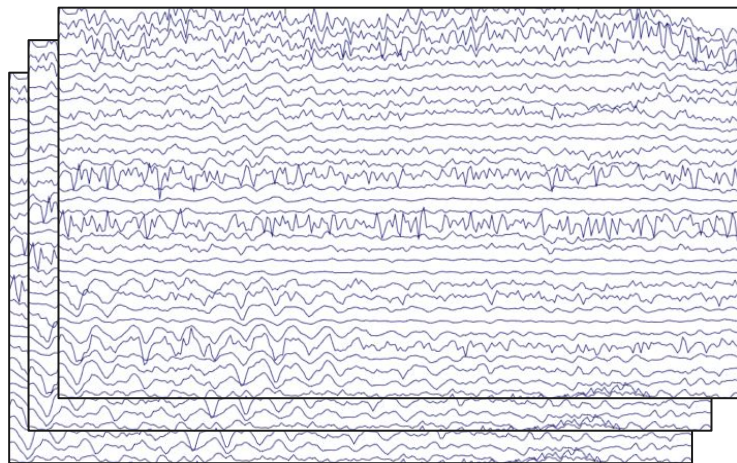
[10] Hight et al. Can. J. Anesth. 2021

# Data collection: awake & anesthetized states

GOAL: Classify brain state as either **MNS-awake** or **MNS-anesthesia** using a BCI.



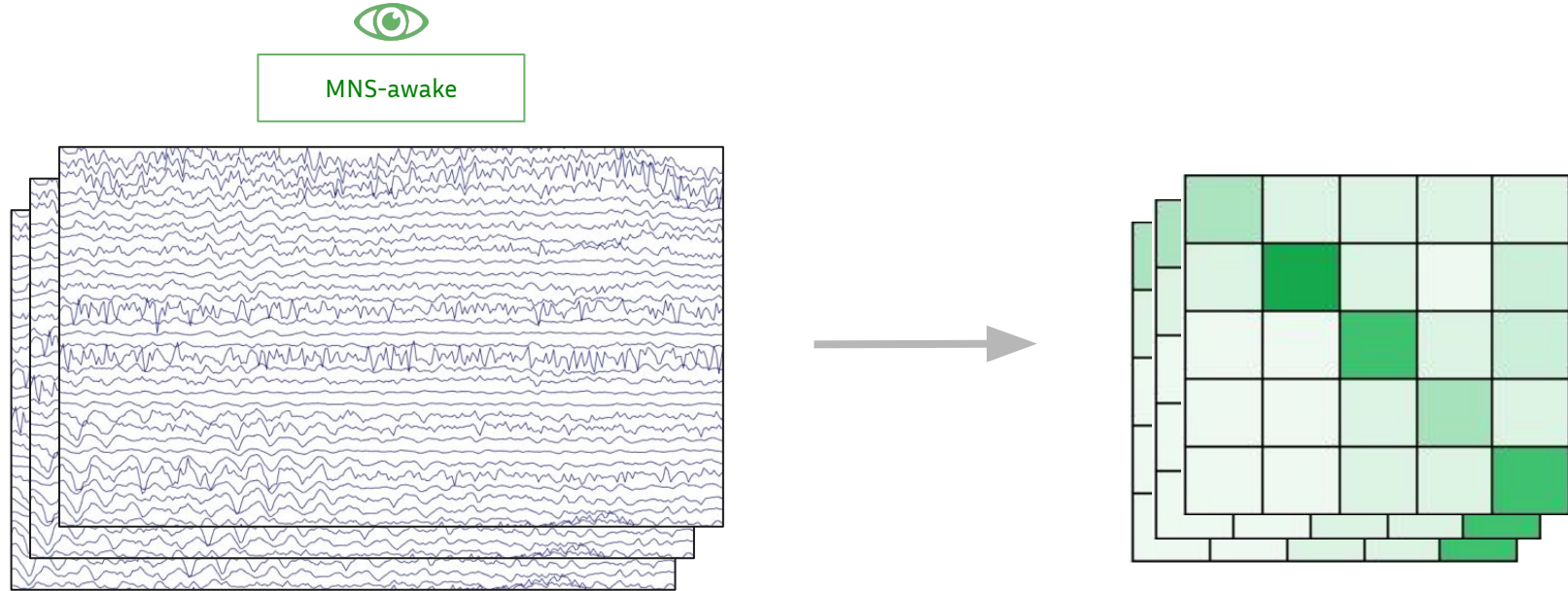
MNS-awake



MNS-anesthesia



# One-Class Riemannian Minimum Distance to the Mean (OCR-MDM)

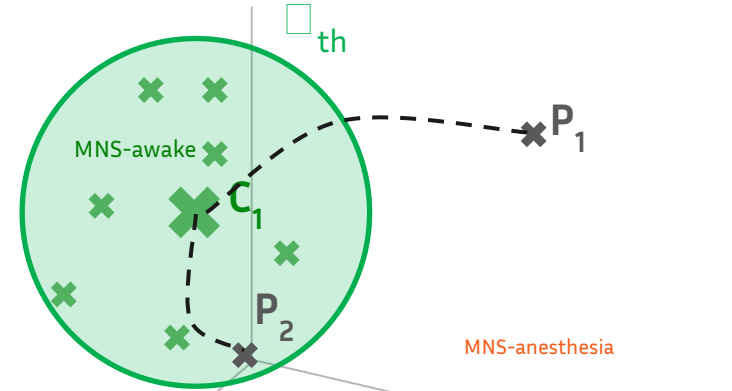
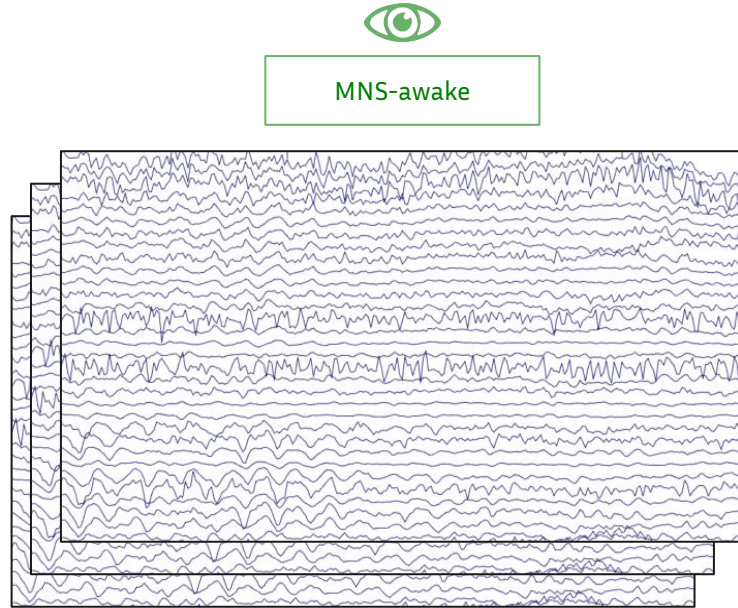


[11] Congedo et al. 8th Inter. BCI Conf. 2019.

[12] Barachant et al. IEEE Trans. Biomed. Eng. 2012.

[13] Marissens Cueva et al. 5th Int. Neuroergonomics Conf, 2024.

# One-Class Riemannian Minimum Distance to the Mean (OCR-MDM)



New covariance matrix  $\mathbf{P}$  of class  $\hat{\mathbf{K}}$ :

$$\hat{\mathbf{K}} = \mathbf{C}_1 \text{ if } \square_{\mathbf{R}}(\mathbf{P}, \mathbf{P}_i) \leq \square_{\text{th}} \text{ else } \mathbf{C}_2$$

with  $\square_{\mathbf{R}}$  the Riemannian distance:

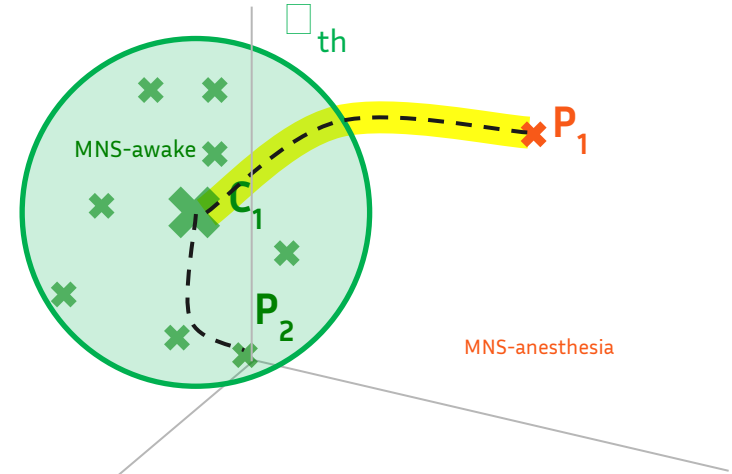
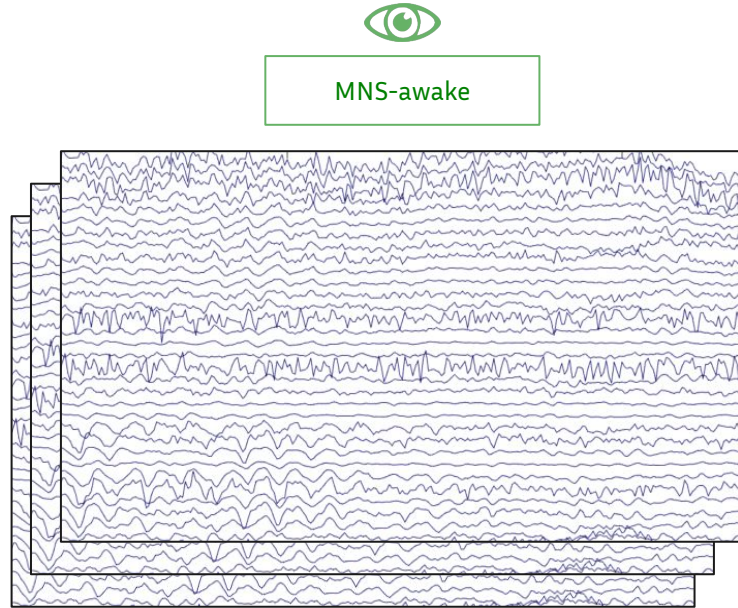
$$\square_{\mathbf{R}}(\mathbf{P}, \mathbf{P}_i) = \|\log(\mathbf{P}^{-1/2} \mathbf{P}_i \mathbf{P}^{-1/2})\|_{\mathbf{F}}$$

[11] Congedo et al. 8th Inter. BCI Conf. 2019.

[12] Barachant et al. IEEE Trans. Biomed. Eng. 2012.

[13] Marissens Cueva et al. 5th Int. Neuroergonomics Conf, 2024.

# One-Class Riemannian Minimum Distance to the Mean (OCR-MDM)



New covariance matrix  $\mathbf{P}$  of class  $\hat{k}$ :

$$\hat{k} = C_1 \text{ if } \square_R(\mathbf{P}, \mathbf{P}_i) \leq \square_{th} \text{ else } C_2$$

with  $\square_R$  the Riemannian distance:

$$\square_R(\mathbf{P}, \mathbf{P}_i) = \|\log(\mathbf{P}^{-1/2} \mathbf{P}_i \mathbf{P}^{-1/2})\|_F$$

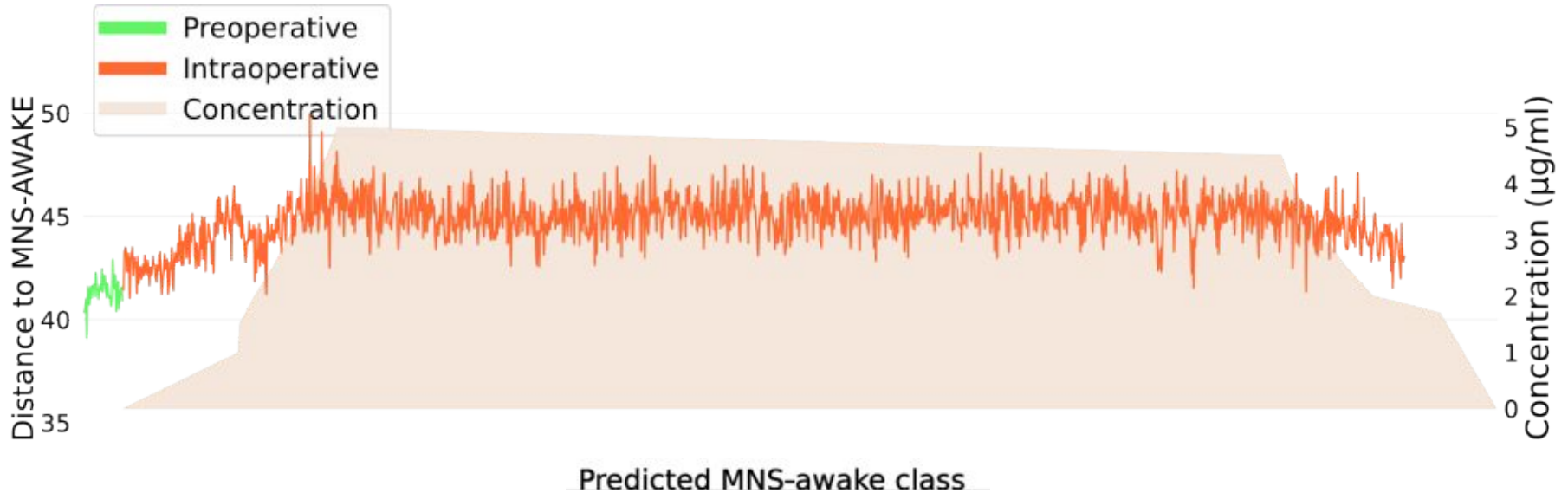
[11] Congedo et al. 8th Inter. BCI Conf. 2019.

[12] Barachant et al. IEEE Trans. Biomed. Eng. 2012.

[13] Marissens Cueva et al. 5th Int. Neuroergonomics Conf, 2024.

# Towards a quantitative anesthesia index

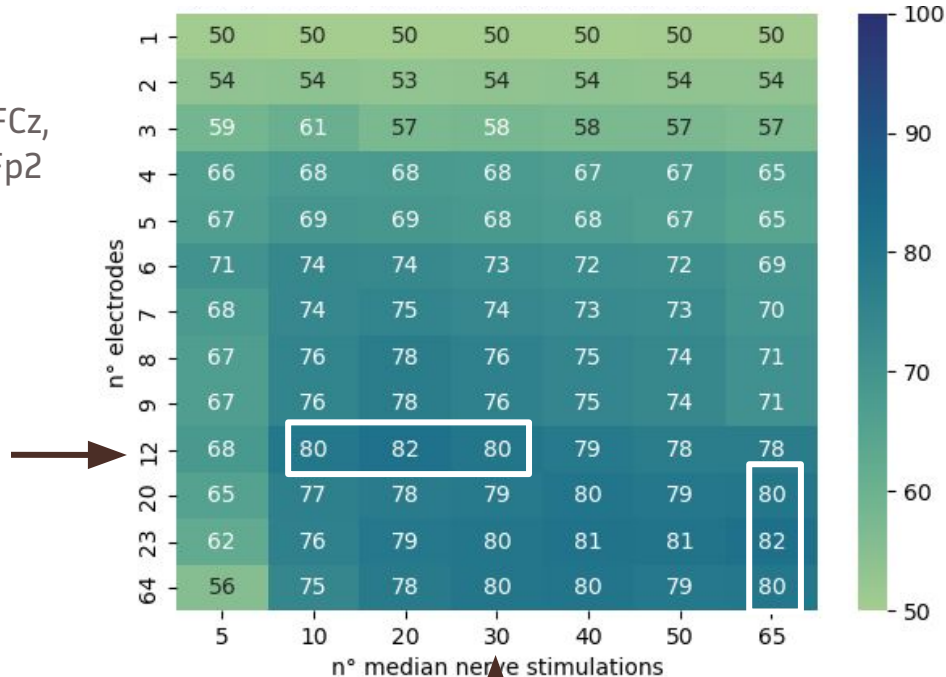
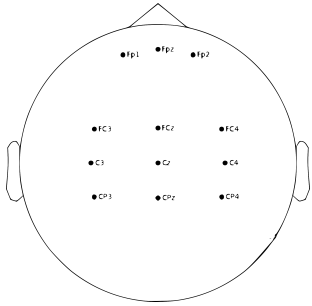
Distance to the **MNS-awake centroid** evolves gradually with the depth of anesthesia



# OCR-MDM performance wrt montage and number of MNS

Test balanced accuracy in the **MNS-awake** vs **MNS-anesthesia** classification:

C3, C4, Cz, FC4, FC3, FCz,  
CP4, CP3, CPz, Fp1, Fpz, Fp2



Movement detection under anesthesia?

# Next steps: movement detection using MNS under anesthesia

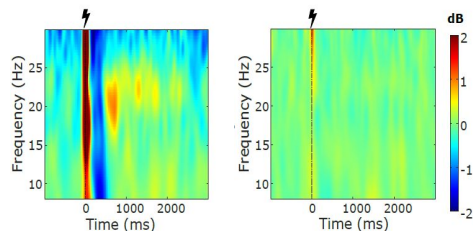
2025 IEEE INTERNATIONAL WORKSHOP ON MACHINE LEARNING FOR SIGNAL PROCESSING, AUG. 31– SEP. 3, 2025, ISTANBUL, TURKEY

## RIEMANNIAN FUSIONS OF EEG-BASED FEATURES FOR MOTOR IMAGERY DETECTION UNDER PROPOFOL SEDATION

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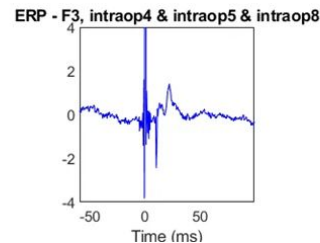
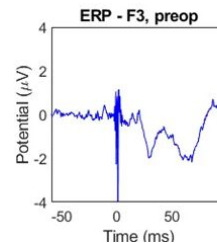
Covariance  
Matrices

Connectivity  
Matrices

NEW EEG FEATURES

Spectral  
Entropy

Somatosensory Evoked  
Potentials



# Acknowledgments



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Université de Lorraine



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Inria Center at the University of Bordeaux

# ANR BCI4IA

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BCI4IA project  
2022-2026



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Alvarez



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# Thank you for your attention!

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